Bracketing a root by bisection

Upper high school · analysis and algorithms · Lesson and method · 12

Prerequisites. Continuity, signs and intervals.

Key point · Ideas and method. If f is continuous on [a,b] and f(a), f(b) have opposite signs, the intermediate value theorem guarantees at least one root in [a,b]. It does not guarantee uniqueness. Compute $m = \frac{a+b}{2}$. If $f(m) = 0$, a root has been found. Otherwise retain the half whose endpoint values have opposite signs. Continuity maintains the guarantee at every step.
After n halvings, the interval width is $\frac{b-a}{2^n}$. Its midpoint approximates a root with error at most half that width. Width is therefore different from midpoint error.

Example 1. For $f(x) = x^2 - 2$ on $[1, 2]$, f is continuous and strictly increasing, so its root is unique. Midpoints 1.5 and then 1.25 give intervals $[1, 1.5]$ and $[1.25, 1.5]$: the left value remains negative and the right value positive.

Activity 1 · Explore. On a number line, draw the successive intervals around $\sqrt{2}$ and label each width.